3.8.26	Setting Up a Local Area Network

AIM:

To design, setup and configure a LAN connecting a components through a network design, switch, assign IP address to all devices. Verify connectivity using ping command and demonstrate resource or data sharing across network.

Equipments Required:

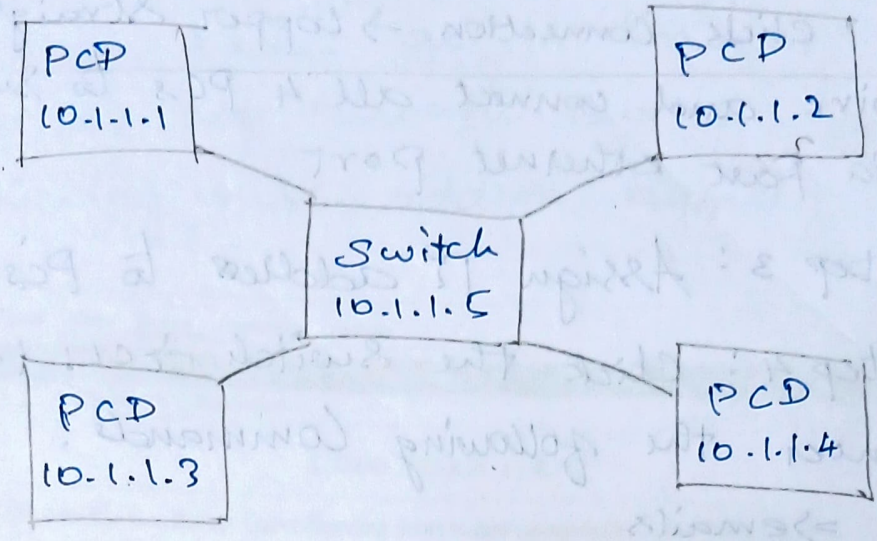
1	Personal Computer	4	Ethernet (NIL) enabled
2	Network Switch	1	8/16/24 port- managed
3	Ethernet Cables (Cat 5e/cat 6)	4	Straight through RJ-45 connectors.

Theory:

*LAN is a network that connects compter & devices within limited area like home, otha

*Star Topology - all end devices are individually connected to cutral switch

Network Diagram:



IP	Address	Configuration	Table
PC0	10.1.1.1	255.0.0.0	Switch Port Fa 0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa 0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa 0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa 0/4
Switch	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation of Cisco Packet Tracket:

Physical hardware was limited, the LAN was also implemented and verified using cisco Packet Tracer Simulation software following the steps below.

Step 1: Place the Device

Open Cisco Packet Tracer. Select 4 PC's onto workspace Select & drag & One switch.

Step 2: Cable the Device

Click connection → copper Straight-wire and connect all 4 PCs to Switch via Fast ethernet port

Step 3: Assign IP address to PCs

Step 4: Click the Switch → CLI tab. Enter the following commands:

=> enable
=> configure terminal
=> interface vlan1
=> ip address 10.1.1.5 255.0.0.0
=> no shutdown
-> exit

Step 5: Verify Connectivity using Ping

Click PC → Desktop → Command Prompt

Use Ping command to test connectivity.

Ping 10.1.1.2

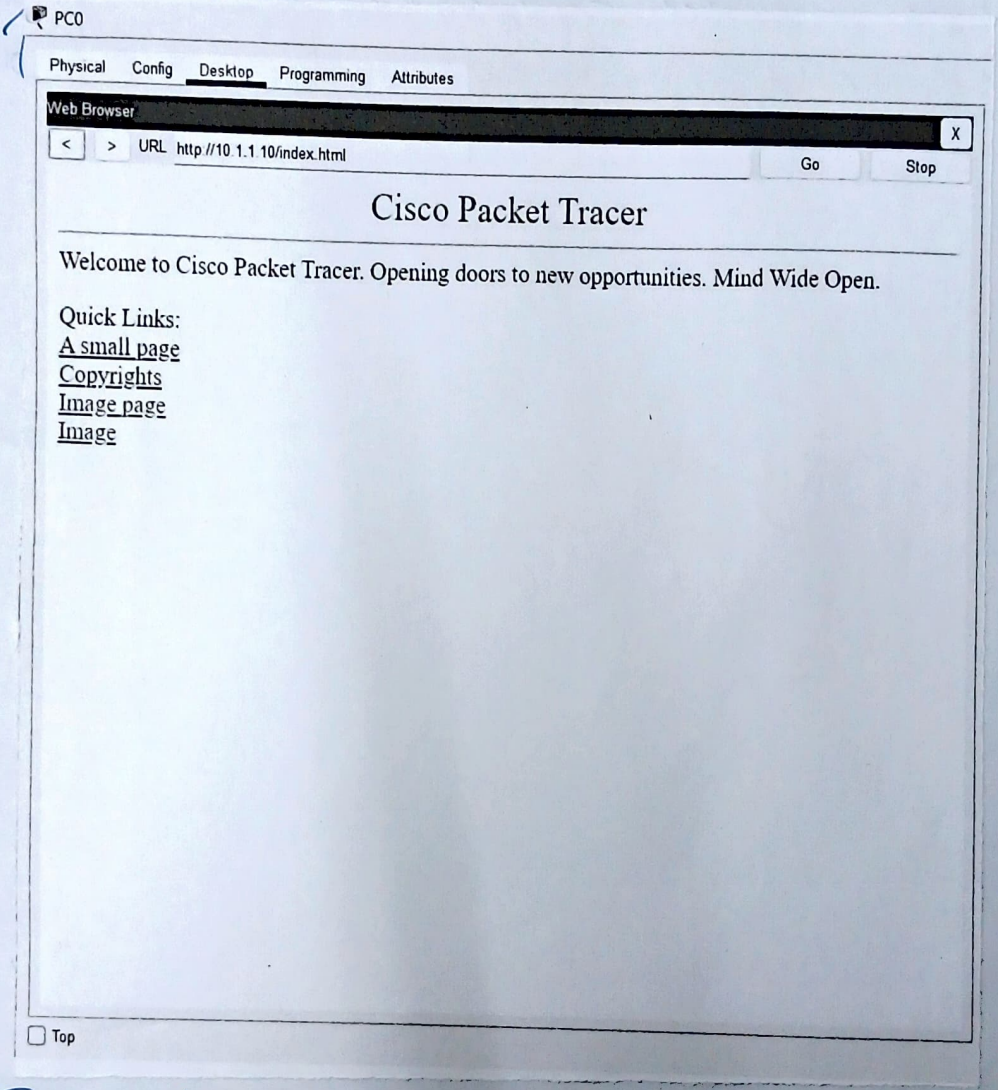
Step 6: Demonstrate Data Sharing

Add a service device and connect it to switch. Assign it ip. Open the servers service.

Step 7: Observe the packet flow in simulation mode.

Switch to simulation mode send a ping request between

two device. Capture the step through packet as its travel hop by hop through switch. Observing mac address based



Result:

Thus, the setting up a local area network using a switch and Ethernet cables has been checked successfully.